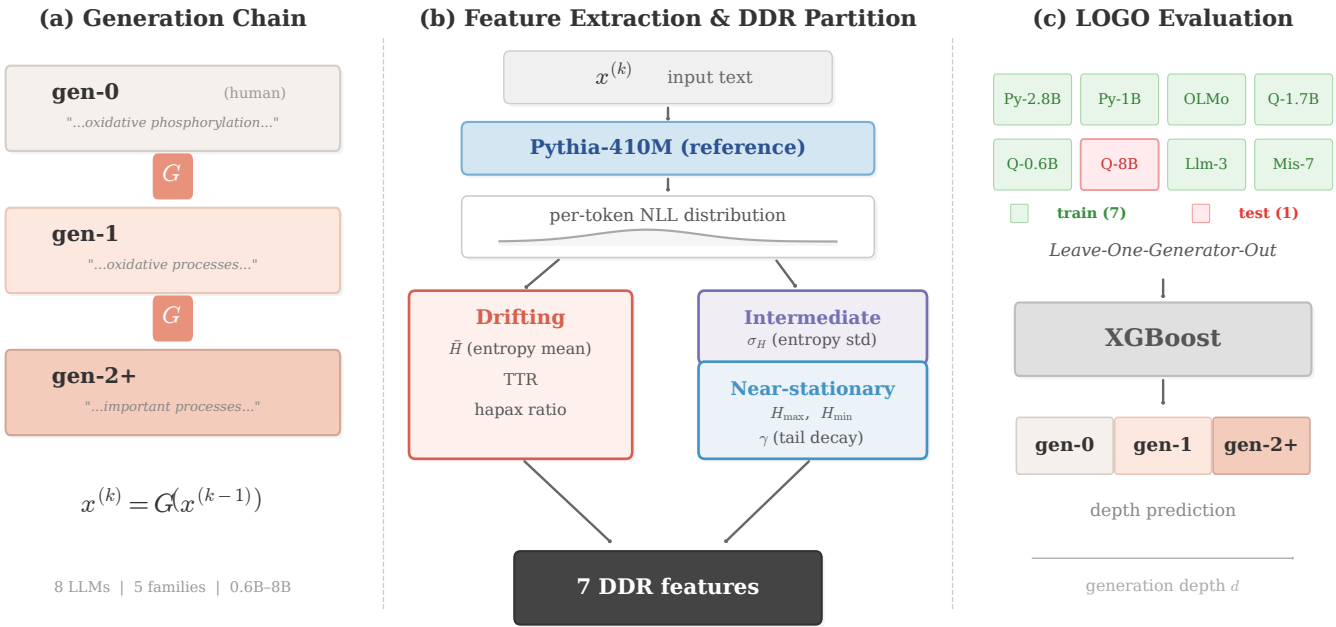
(a) Generation Chain
gen-0
(human)
"...oxidative phosphorylation..."
G
gen-1
"...oxidative processes..."
G
gen-2+
"...important processes..."
$x^{(k)} = G(x^{(k-1)})$
8 LLMs | 5 families | 0.6B–8B
(b) Feature Extraction & DDR Partition
$x^{(k)}$ input text
Pythia-410M (reference)
per-token NLL distribution
Drifting
$\bar{H}$ (entropy mean)
TTR
hapax ratio
Intermediate
σ_H (entropy std)
Near-stationary
$H_{\max}$, $H_{\min}$
γ (tail decay)
7 DDR features
(c) LOGO Evaluation
Py-2.8B
Py-1B
OLMo
Q-1.7B
Q-0.6B
Q-8B
Llm-3
Mis-7
train (7)
test (1)
Leave-One-Generator-Out
XGBoost
gen-0
gen-1
gen-2+
depth prediction
generation depth d